Chapter 7
Ectomycorrhizal Fungal Communities of Edible Ectomycorrhizal Mushrooms

Alessandra Zambonelli, Mirco Iotti, Siham Boutahir, Enrico Lancellotti, Claudia Perini, and Giovanni Pacioni

7.1 Introduction

Ectomycorrhizal (EM) fungi are a wide group of fungi that include around 20,000–25,000 species (Rinaldi et al. 2008). An estimated 1,000 species of EM fungi produce edible fruiting bodies (Hall et al. 2011; Chap. 1). Edible ectomycorrhizal mushrooms (EEMMs) are a quite heterogeneous group of fungi characterized by different reproductive structures either epigeous (mushrooms, puffballs-like, etc.) or hypogeous (truffle-like) that just happen to be edible. As a consequence, they do not form a distinct taxonomic group but are instead spread across many Ascomycota and Basidiomycota genera. Hence, they cannot be expected to have similar niches in soil communities but are instead adapted to different soil conditions (Smith and Read 2008).

In natural habits EM fungi form complex communities characterized by multiple species, and coexistence is governed by competition dynamics (Kennedy 2010) and host availability (Tedersoo et al. 2012). EM fungal competition for host plant carbon resources and soil nutrients is affected by biotic and abiotic factors particularly host plant, soil chemistry, and environmental conditions (Jumpponen and

A. Zambonelli (✉) • M. Iotti • S. Boutahir
Dipartimento di Scienze Agrarie, Università di Bologna, via Fanin 46, 40127 Bologna, Italy
e-mail: alessandr.zambonelli@unibo.it; mirco.iotti2@unibo.it; bsiham2007@yahoo.fr.it

E. Lancellotti
Dipartimento di Protezione delle Piante – sezione di Patologia Vegetale, Università di Sassari, via de Nicola 9, 07100 Sassari, Italy
e-mail: lance@uniss.it

C. Perini
Dipartimento di Scienze Ambientali "G. Sarfatti", Università degli Studi di Siena, via Mattioli 4, 53100 Siena, Italy

G. Pacioni
Dipartimento di Scienze Ambientali, Università dell'Aquila, via Vetoio 1, loc. Coppito, 67100 L'Aquila, Italy

A. Zambonelli and G.M. Bonito (eds.), *Edible Ectomycorrhizal Mushrooms*, Soil Biology 34, DOI 10.1007/978-3-642-33823-6_7,
© Springer-Verlag Berlin Heidelberg 2012

Egerton-Warburton 2005; Kennedy 2010). The soil also includes numerous other microorganisms such as pathogenic and saprobic fungi and bacteria that are known to interact with the EM fungi (Paulitz and Linderman 1991; Cairney and Meharg 2002; Frey-Klett et al. 2007) although these interactions remain poorly investigated (Lindahl et al. 2001; Zambonelli et al. 2009).

The knowledge of EM fungal community composition and dynamics in relation to environmental conditions is an important ecological basis for understanding interactions between different EM fungi. EM competitive relationships can affect survival and spread of EM species in the soil. This is particularly important in EEMM cultivation where a valuable EM species is introduced in the field through the planting of EM infected plants (see Chap. 9). Replacement of the introduced EEMM by native EM fungi is one of the most important causes of truffle cultivation failures (Hall et al. 2007).

This chapter aims to provide a brief review of the approaches applied to the study of EM fungal communities including morphological and molecular methods, as well as resources available on the Web for the identification of EM fungi. Further, we report on recent studies specifically carried out on EEMM communities.

7.2 Methods Used in the Study of EM Fungal Communities

7.2.1 Fruiting Body Surveys

The classical approach in the study of EM fungal communities was simply through collection and identification of fruiting bodies (Arnolds 1991). Fungal species were determined as mycorrhizal through the basis of mycorrhizal synthesis or direct observation of this condition with the host plant in nature (Brundrett 2008; Chap. 1). Such approach tends to underestimate the EM fungal diversity because fruiting body production is sporadic and seasonal; many species have cryptic fruiting, while others may not fructify at all such as some species of Thelephoraceae, Sebacinaceae, and *Cenococcum geophilum* Fr. (Horton and Bruns 2001), all of which have been shown to be common in many edible ectomycorrhizal (EEM) fungal communities (see Sect. 7.3). Moreover, due to the climatic dependence of EM fruiting body production, numerous surveys over many years are necessary to get baseline estimates of fungal diversity (Dahlberg 2001). For these reasons, molecular and morphological analyses of mycorrhizas are now standard methods used to study EM fungal communities.

7.2.2 Morphological Identification of Mycorrhizas

Investigation of the EM fungal communities by morphological characterization (morphotyping) of their mycorrhizas was systematically defined by Agerer (1987–2008, 1991), who provided a large set of morphological characters to unequivocally

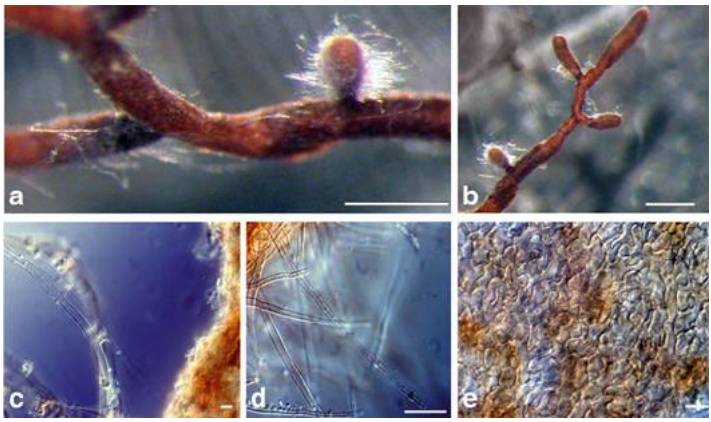

Fig. 7.1 Mycorrhizas of *T. magnatum* obtained by spore inoculation in greenhouse: infected tips (**a** and **b**, bar = 1 mm), external hyphae forming an anastomosis (**c**, bar = 10 μm), awl-shaped cystidia (**d**, bar = 10 μm), epidermoid external mantle (**e**, bar = 10 μm)

describe discrete EM morphotypes. This approach results in a panorama of underground EM fungal diversity. However, using Agerer's methods, only a small fraction of the EM diversity can be attributed to define fungal species with any certainty, and most EM morphotypes remain unidentified. In these cases, the EM species is named using a provisional binomial derived from the genus of the host tree and a characterizing epithet, for example, *Piceirhiza gelatinosa* (Agerer 1991). Studies of EM communities using only morphological methods tend to produce an erroneous estimation of the EM richness. Indeed, multiple species could be included in single morphotypes (e.g. cryptic species), but it is just as likely that a single fungal species produces different morphotypes (Burke et al. 2005). In fact, the morphology of the symbiotic structures formed by an EM fungal species can vary depending on the host plant and the fungal strain, as well as the environmental conditions, tip aging, and phenotypic expression (Martin et al. 2007). For instance, different strains of the EEMM *Tuber borchii* Vittad. were found to form slightly different morphotypes (Giomaro et al. 2000; Sisti et al. 2003), and in the same way the host plant was shown to influence the morphology of the *Tuber brumale* Vittad. mycorrhizas (Giomaro et al. 2002).

Among the EEMMs, morphotyping of mycorrhizas is a commonly used tool for identifying species of truffles in the genus *Tuber*. Truffle mycorrhizas were first described by Zambonelli et al. (1993, 1995), who provided the first key for their identification on *Quercus pubescens* Willd and *Pinus pinea* L. host plants. The most informative characters for *Tuber* mycorrhizas are color, form of cystidia, and shape and dimension of the mantle cells (Fig. 7.1). These characters are used for an online interactive key, called "TuberKey" (Zambonelli et al. 2000b). Even if these keys are useful for the identification of *Tuber* mycorrhizas in commercial seedlings, morphotyping does not always guarantee correct identification of symbiotic truffle species. For example, the mycorrhizas of *Tuber indicum* Cooke and

Massee and *Tuber melanosporum* Vittad. (Zambonelli et al. 1997) have similar cystidia and mantle cell shapes, and in some stages of development, cystidia may be lacking altogether. The same is true for those of *T. borchii* and *Tuber maculatum* Vittad., as well as some other species of *Tuber* (Zambonelli et al. 1999). Morphotyping is a useful technique for discriminating mycorrhizas formed by competitive EM species in truffle orchards and nurseries, such as the "AD" morphotype, described in French for the right angle "angles droits" cystidia cells (Giraud 1988) and more recently identified as *Trichophaea woolhopeia* (Cooke and W. Phillips) through DNA sequencing approaches (Agueda et al. 2008; Rubini et al. 2011).

7.2.3 Molecular Identification of Mycorrhizas

The recent explosion of studies applying PCR-based molecular techniques in fungal identification and genotyping has greatly advanced our understanding of fungal diversity (Peay et al. 2008). The internal transcribed spacer (ITS) rDNA region is now widely used as a DNA barcode for taxonomic identification of ectomycorrhizal fungi (see Chap. 2) and is the foremost marker used to characterize whole fungal communities. The comparison of the ITS sequences obtained from an EM community against those deposited in International Nucleotide Sequence Database (INSD: GenBank, EMBL, and DDBJ) is commonly performed by BLAST best-hit analyses because of its speed and simplicity (Porter and Golding 2011). This approach often allows for the identification of the fungal taxa of interest at genus or even species level resolution, or, at worst, to assign it to one of the main EM lineages (thelephoroid, sebacinoid, agaricoid, etc.). However, there are problems concerning the taxonomic reliability of public sequence databases (Nilsson et al. 2006), and caution is warranted in taxonomic determinations made solely through BLAST searches. For this reason, it is convenient to perform BLAST searches against multiple databases and to adopt a conservative naming approach when the more significant BLAST hits are poorly supported or contrasted. A case of inconsistent reliability of the public sequence database occurs for the genus *Tuber*. While ITS accessions of many distinct species, such as *Tuber magnatum* Pico and *Tuber aestivum* Vittad., appear to be accurately identified, those of other less studied taxa are not. For example, the 107 ITS1-5.8S-ITS2 sequences labeled as *T. borchii* deposited in GenBank (accession date 4 Apr 2012) fall into five different phylogenetic groups (similarity <90 %) that represent distinct species (see also Zambonelli et al. 2010). In another instance, four ITS sequences with a pairwise similarity exceeding 99 % (EU784422, EU784423, EU784424, EU784428) obtained from specimen vouchers of the Kew herbarium (Brock et al. 2009) have mistakenly been accessioned as three different species (*T. borchii*, *Tuber dryophilum* Tul. & C. Tul. and *T. maculatum*). This is in part due to the difficulties of discriminating species of whitish truffles from each other, given the few distinguishing microscopic characters of spores and peridium that have been found useful. Similar cases in GenBank are common for other groups of EM fungi. Effort has been made to annotate reliable sequences in public databases (Tedersoo et al. 2011).

Various criteria have been reported in literature for taxonomic affiliation of morphotypes forming an EM community based on ITS sequence similarity values (Landeweert et al. 2003; Smith et al. 2007) although it is unrealistic to establish a global cut-off value for taxon delimitation that works for all groups of fungi. In fact, the ITS region is not equally variable across the fungi and far-reaching knowledge of genetic variability within each lineage is still needed (Nilsson et al. 2008). The best way to classify species detected on the roots is through the comparison of their ITS sequences to those of the fruiting bodies collected in the same experimental area or from a curated reference databases (Dahlberg 2001; Liu et al. 2011).

The identification of fungal symbionts is routinely done these days through DNA extraction from one or a few colonized root tips, followed by PCR analyses and sequencing. In order to reduce time and costs of molecular identification, "direct PCR"-based method has been developed by Iotti and Zambonelli (2006) and can be applied to the molecular characterization of EEM fungal communities (Peintner et al. 2007; Iotti et al. 2010; Benucci et al. 2011a). This method amplifies the ITS1-5.8S-ITS2 region directly from a small fragment of fungal mantle (0.01–0.02 mm^2 wide) included in the PCR reaction mixture. The fragment of fungal mantle is preferentially excised using fine needles from the distal end of an EM root tip under a stereomicroscope and then transferred directly into the PCR tube. The addition of 20–40 μg of bovine serum albumin (BSA) to the PCR mixture is necessary to scavenge Taq DNA polymerase inhibitors such as phenolic compounds and melanins, which can strongly limit the activity of Taq DNA polymerase (Kreader 1996; Giambernardi et al. 1998; Sejalon-Delmas et al. 2000). Direct PCR techniques bypass the time and expense of DNA isolation and preserve the structure of the mycorrhized tips, which are then available for morphological characterization. This requisite has two primary advantages (1) it allows the description of rare morphotypes, and (2) it can be used to correlate erroneous molecular or morphological data when similar mycorrhizas formed by different fungal species are mixed on the same roots. The risk of amplifying DNA from mycelium or other propagules of nontarget fungal species in the rhizoplane is also drastically reduced. To date only <0.5 % of nontarget fungal species have been amplified from our research staff after analyses of about 500 EM morphotypes collected from different experimental sites. Similar direct PCR method has been also used for identification of truffles and other EEM fruiting bodies (Bonuso et al. 2006; Bonito 2010).

The great number of ITS fungal sequences deposited in GenBank during the last 10 years made possible the design of species-specific primers that are used in simple or multiplex PCR and are able to identify unequivocally mycorrhizas EEMMs (Table 7.1). The use of specific primers coupled with direct PCR approaches allows for a drastic reduction in time and cost for genotyping and routine screening. Moreover, specific primers can be used to detect mycorrhizas or mycelium of target taxa in soils and natural habitats and allow researchers to follow fungal dynamics with the aid of "real-time" PCR (Peintner et al. 2007; Hortal et al. 2008; Zampieri et al. 2010; Suz et al. 2008; de la Varga et al. 2012; Iotti et al. 2012). Specific primers could be a useful tool for assessing the geographical distribution and ecology of *T. aestivum* in countries such as the Czech Republic and Slovakia, where the collection of fruiting bodies is forbidden (Gryndler et al. 2011).

Table 7.1 A set of ITS-specific primers for identification of EEM

EEM species	Specific primers[a]	Sequences	References
Tuber melanosporum	**ITSML**[b]	TGGCCATGTGTCAGATTTAGTA	Rubini et al. (1998)
	ITS1TM[c]	GTATTCCCGAACACAAACCT	Suz et al. (2006)
	ITS2TM[c]	AGACTTGTGACTGATCCAGG	
	T.mel_for	TTGCTTCCACAGGTTAAGTGA	Bonito (2010)
	T.mel_rev	TAAAGTCCGTCGTTCATGC	
T. brumale	**SYLV1**	TCGTCAGTGGTACACAATGT	Douet et al. (2004)
	SYLV2	ATGACAGAACAAATAGAATGTAA	
	ITSB[b]	CAATGTCAGAGCCAATCTAATGC	Rubini et al. (1998)
T. indicum	**IndF1**	ACCTGTGGGAGATCTCCAC	Douet et al. (2004)
	IndF2	GGCCATGTGTCAGATTTACTG	
	IndR	CATAGACTAGCAATTCACTCCTG	
	ITSCHCH[b]	AACAACAGACTTTGTAAAGGGTTG	Rubini et al. (1998)
T. aestivum	**Tu1sekvF**	AGAGCACCAAACCACAG	Gryndler et al. (2011)
	Tu2sekvR	ACCACAGCGTCTACCAA	
	UncI	TGGGCCGCCGAAAACTTG	Mello et al. (2002)
	UncII	CTGACGAGATGCCCCGGA	
T. macrosporum	**Tmacr For**	CGTCGCTCATCAAAGCAGTC	Benucci et al. (2011b)
	Tmacr Rev	CCGCCAGTACCACCAGGAG	
T. rufum	**Ru1f**[d]	TGCTTTCCCAGGTGGTTGG	Iotti et al. (2007)
	Ru2f*[d]	TTGCTTTCCCAGGGAATTGG	
T. magnatum	**Tmag1**[e]	GGATGCGTCTCCGAATCCTGAAT	Amicucci et al. (1998)
	Tmag2	TCGGGCCCTTTCTCAGACTGCTG	
	TSMAGN	GTCACTGAAAACCCACTCACG	Rubini et al. (2001)
	ITSBACK3	TGAGGTCAACCCAGTTGGACAGT	
	P7	TCCTACCAGCAGTCTGAGAAAGGGC	Mello et al. (1999)
	M3[f]	TGAGGTCTACCCAGTTGGGCAGTGG	
	TmgITS1for[c]	GCGTCTCCGAATCCTGAATA	Iotti et al. (2012)
	TmgITS1rev[c]	ACAGTAGTTTTTGGGACTGTGC	
	TmgITS1prob[c]	TGTACCATGCCATGTTGCTT	
T. maculatum	**TmacI**	GACACAGGCTCCCGATAAAACAC	Amicucci et al. (1998)
	TmacII	CAGCAGCACTGATAGCCCCG	
	rTmacII[e]	CGGGGCTATCAGTGCTGCTG	Amicucci et al. (2000)
T. borchii	**TboI**	TGTATGGGATGCCCTATCGGACT	Amicucci et al. (1998)
	TboII	CTATTACCACGGTCAACTTC	
	rTboII[e]	GAAGTTGACCGTGGTAATAG	Amicucci et al. (2000)
	TBA	TGCCCTATCGGACTCCCAAG	Mello et al. (1999)
	TBB	GCTCAGAACATGACTTGGAG	
T. puberulum	**TpuI**[e]	TCTGTTACCAGGGTCCACATT	Amicucci et al. (1998)
	TpuII	GGCTTCTGGGTTGAGGTGTTT	

(continued)

Table 7.1 (continued)

EEM species	Specific primers[a]	Sequences	References
T. dryophilum	**TdryI**	ATCGGGCTCCCAAGCAAAACA	Amicucci et al. (1998)
	TdryII	TCTACTACCATGGTTCACTTT	
Boletus edulis	**BE1**	CATTATCGAGTTAGACCGGGAAG	Lian et al. (2008)
	BE-2	CCATGCCCTCGAGATCAGATC	
	Bedu1F	ATGGAGGAGTCAAGGCTGTC	Mello et al. (2006)
	Bedu2R	TAGATTAGAAGCGATTCACT	
B. pinophilus	**Bpin1F**	GGGAAAATGGACAAGGACCC	Mello et al. (2006)
	Bpin1R	AGTCAGTCTCACGACCGACTTT	
B. aereus	**Baer2F**	GAAGCCGTTTCCTCGGACTCC	Mello et al. (2006)
	Baer1R	TGAGGCTTAGGGTTCGAGTC	
B. aestivalis	**Baes1F**	AAGGGGTTTCCTCGGACTCT	Mello et al. (2006)
	Baes2R	AATTGGAAGCGATTCACCGCW	
B. violaceofuscus	**Bvio1F**	CACACTTTGTCGCATGTCCAT	Mello et al. (2006)
	Bvio1R	CCACCATCGTGTGCTTGTGT	
Lactarius deliciosus	**FWD-Ldel**[c]	TTGACGCAAAAGTCGTGCAC	Parladé et al. (2007)
	RVS-Ldel[c]	ATCGGTTCGATCCCAAAAGG	
	STQ Ldel[c]	TCTCGCATAAAATCCA	
Rhizopogon roseolus	**FWD-Rro**[c]	TCGACTTTGCGCGACAAG	Hortal et al. (2008)
	RVS-Rro[c]	CATGCGCTTCAGCAAACG	
	STQ-Rro[c]	ATCATTATCACGCCGAAAG	

[a]Forward primer is in bold. Probes are in italic
[b]In multiplex PCR with ITS4LNG (TGATATGCTTAAGTTCAGCGGG) as reverse primer
[c]In real-time PCR
[d]ITS4 as reverse primer
[e]In multiplex PCR with ITS4 as reverse primer
[f]In multiplex PCR with ITS1 as forward primer

7.2.4 Metagenomics

Metagenomics can be defined as "the application of modern genomic techniques to the study of communities of microbial organisms directly in their natural environments, bypassing the need for isolation and lab cultivation of individual species" (Chen and Pachter 2005). Metagenomics is based on the analysis of nucleic acids (DNA or RNA) extracted directly from the environment. Metagenomics can be used to study soil fungal communities and to this aim several approaches are suitable. Polymerase chain reaction-temperature gradient gel electrophoresis/denaturing gradient gel electrophoresis (PCR-TGGE/DGGE), cloning and sequencing of rDNA amplicons and 454 amplicon pyrosequencing have been used to elucidate the composition and diversity of EEM communities (Peintner et al. 2007; Napoli et al. 2010; Mello et al. 2011). An ever-increasing number of studies on EM communities are under way using next-generation sequencing approaches (454 and Illumina

platforms), which provide more complete community characterization and previously unimaginable amount of sequence data. However, quality control measures (e.g. denoising, chimera checking) and reference databases continue to be improved, and caution should be taken when interpreting results given inherent biases of all PCR-based approaches (Kunin et al. 2010; Tedersoo et al. 2010).

Another promising high-throughput molecular tool for studying fungal communities is the microarray technique. PhyloChip design is based on the use of several thousand oligonucleotide probes derived from DNA barcodes to allow the rapid and parallel identification of a high number of species from many processed samples. Various PhyloChips have been designed for detecting target fungal species, but phyloarrays have been poorly applied for characterization of EM communities (Reich et al. 2009), and probes had been designed only for a low number of EEMM species (Bruns and Gardes 1993; El Karkouri et al. 2007). Nonetheless, new and custom arrays and probes are available, including those targeting functional genes, and small-scale PhyloChips are a promising time-saving and cheap approach for monitoring specific fungal species over years and/or space (Reich et al. 2009). However, one major limitation of phyloarrays is that they can only detect known taxa and genes; thus, they are not suitable for discovery applications.

7.2.5 Ectomycorrhizal Database

Recently a number of free online databases have been created to facilitate identification of ectomycorrhizas. DEEMY (http://www.deemy.de) (Rambold and Agerer 1997) and UNITE (http://unite.ut.ee/index.php) (Kõljalg et al. 2005) are the most widely used tools for morphological and genetic data gathering, respectively. Together, UNITE and GenBank are the most important genetic database available for characterizing ectomycorrhizal fungi. Other useful Web databases for research on ectomycorrhizas have been created by (a) Canadian Forestry Service (CFS), which hosts an image database (POE) and a database containing brief descriptions of the ectomycorrhizas (EDD) (http://www.pfc.cfs.nrcan.gc.ca/biodiversity/bcern/index_e.html) and (b) University of Wisconsin (USA) that hosts the "Database of the Chestnut Mycorrhizae," which is a collection of ITS region sequences of EM fungal species in association with plants of the *Castanea* genus. However, these tools are limited and do not give information pertaining to the community to which given EM fungi belong or concerning the role of different species in a given community. Moreover, these databases do not supply exhaustive information on the ecology and the geographical distribution of the fungal species.

Another new Web tool worth mentioning is eMyCo "*e*cto*my*corrhizal *co*mmunity database," which collects and analyzes data about EM communities (Lancellotti et al. 2012). This database has been developed specifically to assist researchers working on the ecology of the EM fungi. eMyCo is composed of independent surveys, each characterizing a single EM community at a certain time and place. For each

survey, site description (host community, soil, geographic localization), survey methodology and operational taxonomic units (OTUs) characterized by a simplified set of morphological features of their mycorrhizas and the ITS sequences of the fungal species are maintained. Abundance and frequency values of each OTU in a community are also provided. The values of OTU richness, of the Shannon (Shannon and Weaver 1949) and Pielou (Pielou 1969) indices, together with a chart representing rank-abundance curve (Whittaker 1965), are calculated automatically. In this way, eMyCo allows the use of data concerning ecological parameters in a coherent way to compare EM communities.

7.3 Studies of EEM Fungal Communities

Reported here is an overview of important studies on EEM fungal communities, with an emphasis on the most important advances concerning community composition and dynamics.

7.3.1 Ectomycorrhizal Communities of "Porcini"

Only a few studies have been conducted on *Boletus edulis sensu lato* (porcini) EM communities. Using morphological and molecular identification, Peintner et al. (2007) surveyed EM, saprobic and pathogenic fungi in an area located near Parma (Italy) that is productive with *B. edulis* s.l. Thirty-nine (39) EM fungi were identified on root tips, whereas 40 fungal species were found in the soil by cloning and sequencing. However, the overlap between above- and below-ground fungal communities was very low, and *Boletus* mycelia were rare and scattered. DNA from most species within the *B. edulis* s.l. group (*Boletus edulis* Bull., *Boletus pinophilus* Pilát and Dermek and *Boletus aereus* Bull.) appears to be poorly represented adjacent to fruiting bodies (Peintner et al. 2007). Only *Boletus aestivalis* (Paulet) Fr. (syn. *Boletus reticulatus* Schaeff.) ectomycorrhizas and soil mycelia were detected abundantly just below their fruiting bodies (Peintner et al. 2007). A recent study performed using real-time PCR confirms that *B. edulis* productivity (in terms of fruiting bodies) was not correlated either with the concentration of soil mycelium or with the presence or abundance of ectomycorrhizas (de la Varga et al. 2012).

7.3.2 EM Communities of T. melanosporum

T. melanosporum EM communities have been studied with much attention to the persistence of its mycorrhizas and its competitiveness with resident fungal species when infected plants are transplanted for establishing truffle orchards. Many of

these studies were carried out in France, in Spain, and in Italy during the last 30 years using morphological methods (Giraud 1988; Granetti and Baciarelli Falini 1997; de Miguel and Sáez 2005). These studies identified a number of EM species commonly found in truffle orchard, which appear to be competitive with *T. melanosporum*. In particular, two other *Tuber* species, *T. brumale* and *T. aestivum*, appear to be quite competitive with *T. melanosporum* in many conditions (see Chap. 10). Other EM fungi considered as highly competitive and agressive with *T. melanosporum* belong to the *T. woolhopeia* species complex (Agueda et al. 2008; Rubini et al. 2011), which have been known for decades as forming the AD morphotype.

In order to better understand the ecology of *T. melanosporum* and its relationships with other EM fungi, a number of field studies in natural truffières have been carried out in zones that produce truffles and others that do not produce truffles. Productive zones usually correspond with the brûlé (burnt area), an area of soil around the host plant where the growth of grasses is drastically reduced by substance (probably volatile organic compounds—VOCs) produced by the truffle mycelium (Pacioni 1991; Splivallo et al. 2007; Streiblová et al. 2012). Results from quantitative PCR indicates that mycorrhizas and mycelium of *T. melanosporum* are particularly abundant inside the brûlé and decrease beyond it (Garcia-Barreda and Reyna 2011; Suz et al. 2008). The high abundance of *T. melanosporum* inside the brûlé appears to lower the overall species richness of EM fungi and fungal soil communities (Pacioni 1991; Garcia-Barreda and Reyna 2011; Napoli et al. 2008; Mello et al. 2011; Belfiori et al. 2012). Also the composition of the fungal communities is affected by *T. melanosporum* inside of the brûlé. The studies of Mello et al. (2011) carried out using pyrosequencing in truffle ground located in France showed that the number of Ascomycota, which were found to be the dominant fungal phylum in the studied truffle ground, is more abundant inside the brûlé than outside. In contrast, the number of Basidiomycota increases outside the brûlé confirming the previous results obtained by Napoli et al. (2010) by DGGE technique. Recent studies on soil fungal biodiversity carried out in central Italy in *T. melanosporum* natural and cultivated truffle grounds (Belfiori et al. 2012) highlighted only few EM OTUs such as Basidiomycetes of the genus *Hymenogaster* and members of Thelephoraceae are in common with the French truffle grounds studied by Napoli et al. (2010) and Mello et al. (2011).

7.3.3 EM Communities of T. aestivum

T. aestivum EM fungal communities have mostly been investigated in truffle orchards. In one of the first studies, carried out in a 5-year-old *T. aestivum* orchard using morphological approaches, it was possible to detect 15 nontarget morphotypes, among which were *T. brumale* and *Hymenogaster citrinus* Vittad. (Zambonelli et al. 2005). In this study, the effect of different mulching materials on the EM community was compared, and it was shown that mulch type significantly influences the degree

of EM colonization in the upper 0–15 cm of soil. The type of mulching material appears to have effected ectomycorrhizal colonization through its impact on soil temperature and moisture. The decreased soil temperatures in summer and the increased soil moisture caused by straw and black mulching cloth favor contaminant EM fungi in the studied environmental conditions. Similar results were obtained in Spain by Etayo and De Miguel (2001).

In one study, *T. aestivum* EM communities were studied in a productive 24-year-old orchard in Italy using molecular methods and by sequencing the ITS regions. In this study, 29 EM taxa were identified (Benucci et al. 2011a). The most abundant EM species was *Tricholoma scalpturatum* (Fr.) Quél., a species that has also been detected in a *T. magnatum* natural truffière in northern Italy (Iotti and Zambonelli 2006). Similar to other truffle communities, multiple *Tuber* species are present (e.g. *T. rufum* Pico, *T. brumale* and *Tuber rapaeodorum* Tul. & C. Tul.), and the Thelephoraceae shows high diversity (Benucci et al. 2011a). Other EM species detected belonged to Pyronemataceae, Sebacinaceae, Inocybaceae, families showing a similar composition of truffle fungal communities in other studies (e.g. Bonito et al. 2011). *T. aestivum* mycorrhizas were abundant and showed a certain host preference for hornbeam (*Carpinus betulus*), rather than hazelnut (*Corylus avellana*) (Benucci et al. 2011a).

7.3.4 EM Communities of T. borchii

There are few studies on *T. borchii* EM communities in orchards compared to *T. melanosporum* because its cultivation is quite recent. In recent years, however, *T. borchii* cultivation has become popular in many countries (Hall et al. 2007). In a pioneering study conducted in a *P. pinea–T. borchii* orchard, it was shown that *T. borchii* ectomycorrhizas were widespread in the truffiére four years after planting, when the orchard began to produce *T. borchii* fruiting bodies (Zambonelli et al. 2000a). However, the number of *T. borchii* ectomycorrhizas drastically decreased 12 years after planting when the canopies closed, which suggests that *T. borchii* is an early stage fungus and that its ectomycorrhizas are distributed in the youngest part of root system (Zambonelli et al. 2000a). A study on *T. borchii* EM communities carried out in natural truffières confirmed that *T. borchii* mycorrhizas are very abundant in productive points but are present also one meter far from them (Iotti et al. 2010). Seventy ectomycorrhizal taxa were identified, many of which were rare. *T. borchii* dominated belowground, forming 20 % of ectomycorrhizas. Thelephoraceae, Inocybaceae, and Sebacinaceae were the other main EM species. Species composition was markedly affected by the host plant (oak or pine), yet community structure and composition was also influenced by the location the soil cores were collected from. The high presence of *T. borchii* in productive points affected the composition of EM community structure in a similar way as was observed in *T. melanosporum*- and *T. aestivum*-dominated communities (Benucci et al. 2011a; Napoli et al. 2010). In *T. borchii* EM communities, other *Tuber* species

were found sharing the same habitat, in particular, *Tuber dryophilum* Tul. and C. Tul, an inferior edible truffle morphologically similar to *T. borchii* but without commercial value. However, its mycorrhizas were never found together with those of *T. borchii* showing the two species differentiate in the micro-niche colonization as suggested for other EM fungi in *Pinus* forests (Iwanski and Rudawska 2007). Also, at this site multiple cryptic species of *T. borchii* were present (Bonuso et al. 2010) but their mycorrhizas were never found mixed together (Iotti, unpublished data).

7.3.5 EM Communities of T. magnatum

T. magnatum, the Italian white truffle, is the most expensive EEMM species. Nevertheless, with all of the attention that scientists have given this species, its cultivation is still not feasible and only a few truffle orchards are able to sporadically produce fruiting bodies (Hall et al. 2007). Even if methods have been developed to produce *T. magnatum*-infected plants (Mello et al. 2001; Rubini et al. 2001) (Fig. 7.1), mycorrhizas of this truffle seem to disappear when the plants are transplanted in the field (Hall et al. 2007). Also, in natural productive *T. magnatum* orchards, mycorrhizas appear to be absent (Di Massimo et al. 2010) or quite rare (Bertini et al. 2006; Murat et al. 2005). In contrast, other species of *Tuber* such as *T. rufum*, *T. brumale*, *T. maculatum*, and *T. borchii* (Murat et al. 2005; Bertini et al. 2006; Iotti and Zambonelli 2006) are frequent EM fungi on host roots in natural *T. magnatum* truffières. For instance, soil analyses using specific *Tuber* primers revealed a surprisingly high *Tuber* diversity in natural *T. magnatum* truffières with more than five different *Tuber* spp. (Zampieri et al. 2010). Soil analyses using *T. magnatum*-specific primers designed on β-tubulin gene by Zampieri et al. (2010) or more recent real-time PCR analysis using specific ITS *T. magnatum* primers (Iotti et al. 2012) revealed that *T. magnatum* mycelium grows extensively in the soil of the truffière, not only in the productive point but also forming large patches. Soil analysis using DGGE technique carried out in productive and in not productive points of a single *T. magnatum* natural truffière found some EM fungi such as *C. geophilum* and *Clavulina cinerea* (Bull.) J. Schröt. present only in nonproductive points that may negatively influence *T. magnatum* development (Mello et al. 2010).

Indeed, the EM communities of *T. magnatum* natural truffières located in different Italian forest environments are under investigation in order to verify common traits and differences. About 130 EM fungal OTUs have been identified molecularly on the roots collected from four experimental sites, and more than half of them belonged to thelephoroid and sebacinoid taxa. Among the truffle species, *T. rufum* mycorrhizas were the most frequent and abundant. Preliminary results obtained from the samples collected in *T. magnatum* fruiting points revealed different fungal community structures between soil and root EM communities (Fig. 7.2). Only 19 on 45 OTUs were present both as free-living mycelium and as ectomycorrhizas (particularly *Inocybe* spp. and *Sebacina* spp.). Members of

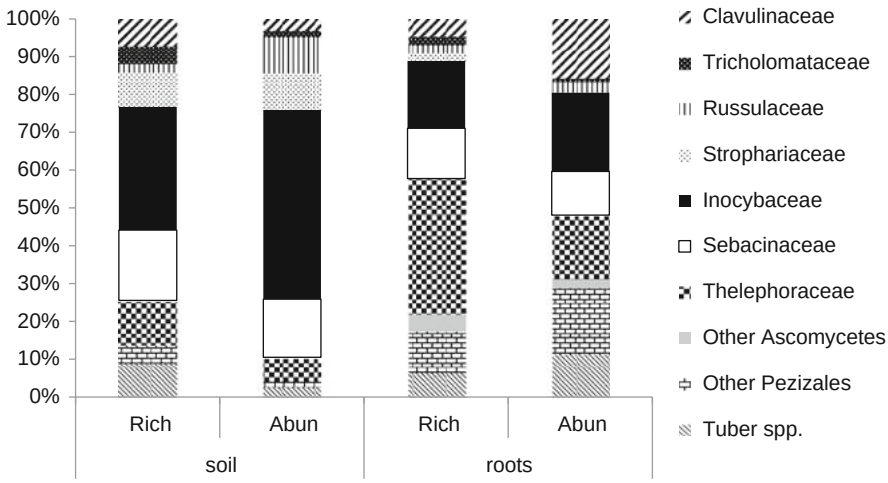

Fig. 7.2 Abundance of the fungal families of soil (**a**) and root (**b**) EM communities present in the fruiting points of two *T. magnatum* natural truffieres located in Tuscany (Montaione, FI) and Emilia Romagna (Argenta, FE)

Inocybe were relatively more abundant in soil than on the roots, whereas ascomycetes, Clavulinaceae, and thelephoroid fungi prefer the symbiotic state (Iotti, unpublished data).

7.4 Conclusions

In cultivating EEMMs, competition from resident EM fungi is one of the biggest problems, particularly in the initial few years after planting when the introduced species is vulnerable to being replaced. This emphasizes the need to better understand the drivers and dynamics of soil EM communities and the environmental factors that favor micro-niche colonization by EEMMs. The application of molecular techniques for taxonomic affiliation has dramatically changed microbial ecology so that it is now possible to investigate the composition and interactions among microbial communities in soil, on the roots, and above-ground, providing a more complete and precise scenario of their structural composition and dynamics.

Through these studies, basic characteristics of the studied EEM fungal communities have emerged:

- Most of EEMMs dominate the EM fungal communities in productive areas (like *T. borchii*, *T. melanosporum*, and *T. aestivum*). These species affect the composition of EM communities reducing richness and abundance of EM species (Iotti et al. 2010; Belfiori et al. 2012).

- Other species like *B. edulis* or *T. magnatum* have a quantity of mycorrhizas in the soil that does not appear to be correlated to the fruiting body productivity; the mycelium of these species in the soil is scattered in productive areas for porcini and widespread in large patches in *T. magnatum* truffières.
- It is common that phylogenetically related species shared the same EM communities, such as some species of *B. edulis* complex, *T. borchii* and *T. dryophilum* and *T. melanosporum* and *T. brumale*. However, most of these species tend to colonize different niches and may coexist within the same habitat. It would be useful to better understand the micro-environmental conditions (like shown in Chap. 6) in micro-niches where the most valuable species are present.

Although our knowledge of the composition EM communities and of soil microbial communities has increased in the past several years, more studies are needed if we are to better understand the ecological variables driving the dynamics of EEM fungal communities, the biotic and abiotic factors affecting synergic or competitive relationships and the fruiting body formation and development. Future studies should take into account the capacity of some EEMMs to form endomycorrhizas with different host species or particular environmental conditions. For instance, it is well known that the EEMMs of the genus *Terfezia* (the desert truffles) are able to form ectomycorrhizas or endomycorrhizas depending on the growth conditions (Fortas and Chevalier 1992; Chap. 14). Moreover, it was recently found that some species of the genus *Tuber* (*T. borchii* and *T. aestivum*) are able to thrive inside host cells of some Orchidaceae (Selosse et al. 2004; Illyés et al. 2010).

These studies, together with advances in genomics (Chap. 21), may be able to solve the mysteries of *T. magnatum* soil biology and ecology and provide the scientific principles to enable its cultivation.

Acknowledgment The authors would like to thank Dr Ian Hall for the critical reading of the manuscript. The work on real-time PCR for detecting and quantifying *T. magnatum* soil mycelium was financially supported by the Tuscany, Emilia Romagna, Abruzzo and Molise regions (project MAGNATUM—Monitoraggio delle Attività di Gestione delle tartufaie Naturali di Tuber Magnatum). The project MAGNATUM was coordinated by ARSIA (Agenzia Regionale per lo Sviluppo e L'Innovazione nel settore Agricolo-forestale) of Tuscany region.

References

Agerer R (1987–2008) Colour atlas of ectomycorrhizae. 1st–14th delivery. Einhorn, Schwäbisch Gmünd

Agerer R (1991) Characterization of ectomycorrhiza. In: Norris JR, Read DJ, Varma AK (eds) Techniques for the study of mycorrhiza. Methods Microbiol 23:25–73

Agueda B, Agerer R, De Miguel AM, Parlade J (2008) *Querciriza quadratum* + *Quercus ilex* L. Subsp ballota (Scop.) Desf. Samp. In: Agerer R, Danielson RM, Egli S, Ingleby K, Luoma D, Treu R (eds) Descriptions of ectomycorrhizae, vol 12. Einhorn, Schwäbisch Gmünd

Amicucci A, Zambonelli A, Giomaro G, Potenza L, Stocchi V (1998) Identification of ectomycorrhizal fungi of the genus *Tuber* by species-specific ITS primers. Mol Ecol 7:273–277. doi:10.1046/j.1365-294X.1998.00357.x

Amicucci A, Guidi C, Zambonelli A, Potenza L, Socchi V (2000) Multiplex PCR for the identification of white *Tuber* species. FEMS Microbiol Lett 189:265–269. doi:10.1111/j.1574-6968.2000.tb09241.x

Arnolds E (1991) Decline of ectomycorrhizal fungi in Europe. Agric Ecosyst Environ 35:209–244. doi:10.1016/0167-8809(91)90052-Y

Belfiori B, Riccioni C, Tempesta S, Pasqualetti M, Paolocci F, Rubini A (2012) Comparison of ectomycorrhizal communities in natural and cultivated *Tuber melanosporum* truffle grounds. FEMS Microbiol Ecol. doi:10.1111/j.1574-6941.2012.01379.x

Benucci GMN, Raggi L, Albertini E, Grebenc T, Bencivenga M, Falcinelli M, Di Massimo G (2011a) Ectomycorrhizal communities in a productive *Tuber aestivum* Vittad. orchard: composition, host influence and species replacement. FEMS Microbiol Ecol 76:170–184. doi:10.1111/j.1574-6941.2010.01039.x

Benucci GMN, Raggi L, di Massimo G, Baciarelli-Falini L, Bencivenga M, Falcinelli M, Albertini E (2011b) Species-specific primers for the identification of the ectomycorrhizal fungus *Tuber macrosporum* Vittad. Mol Ecol Resour 11:378–381. doi:10.1111/j.1755-0998.2010.02915.x

Bertini L, Rossi I, Zambonelli A, Amicucci A, Sacchi A, Cecchini M, Gregori G, Stocchi V (2006) Molecular identification of *Tuber magnatum* ectomycorrhizae in the field. Microbiol Res 161:59–64. doi:10.1016/j.micres.2005.06.003

Bonito G (2010) Fast DNA-based identification of the black truffle *Tuber melanosporum* with direct PCR and species-specific primers. FEMS Microbiol Lett 301:171–175. doi:10.1111/j.1574-6968.2009.01812.x

Bonito G, Brenneman T, Vilgalys R (2011) Ectomycorrhizal fungal diversity in orchards of cultivated pecan (*Carya illinoinensis*; Juglandaceae). Mycorrhiza 21:601–612. doi:10.1007/s00572-011-0368-0

Bonuso E, Iotti M, Macrì A, Zambonelli A (2006) Approccio innovativo per l'identificazione molecolare di funghi filamentosi. Micol Ital 35:32–40

Bonuso E, Zambonelli A, Bergemann SE, Iotti M, Garbelotto M (2010) Multilocus phylogenetic and coalescent analyses identify two cryptic species in the Italian bianchetto truffle, *Tuber borchii* Vittad. Conserv Genet 11:1453–1466. doi:10.1007/s10592-009-9972-3

Brock PM, Dörin H, Bidartondo MI (2009) How to know unknown fungi: the role of a herbarium. New Phytol 181:719–724. doi:10.1111/j.1469-8137.2008.02703.x

Brundrett M (2008) Mycorrhizal associations: the Web resource, Sect 9. Ectomycorrhizal fungi. List of fungi. http://mycorrhizas.info/ecmf.html#list

Bruns TD, Gardes M (1993) Molecular tools for the identification of ectomycorrhizal fungi-taxon specific oligonucleotide probes for suilloid fungi. Mol Ecol 2:233–242. doi:10.1016/j.soilbio.2005.01.028

Burke DJ, Martin KJ, Rygiewicz PT, Topa MA (2005) Ectomycorrhizal fungi identification in single and pooled root samples: terminal restriction fragment length polymorphism (TRFLP) and morphotyping compared. Soil Biol Biochem 37:1683–1694. doi:10.1016/j.soilbio.2005.01.028

Cairney JWG, Meharg AA (2002) Interactions between ectomycorrhizal fungi and soil saprotrophs: implications for decomposition of organic matter in soils and degradation of organic pollutants in the rhizosphere. Can J Bot 80:803–809. doi:10.1139/b02-072

Chen K, Pachter L (2005) Bioinformatics for whole-genome shotgun sequencing of microbial communities. PLoS Comput Biol 1:e24. doi:10.1371/journal.pcbi.0010024

Dahlberg A (2001) Community ecology of ectomycorrhizal fungi: an advancing interdisciplinary field. New Phytol 150:555–562. doi:10.1046/j.1469-8137.2001.00142.x

de la Varga H, Águeda B, Martínez-Peña F, Parladé J, Pera J (2012) Quantification of extraradical soil mycelium and ectomycorrhizas of *Boletus edulis* in a Scots pine forest with variable sporocarp productivity. Mycorrhiza 22:59–68. doi:10.1007/s00572-011-0382-2

de Miguel AM, Sáez R (2005) Algunas micorrizas competidoras de plantaciones truferas. Pub Biol Universidad de Navarra Serie Botánica 16:1–18

Di Massimo G, Benucci GMN, Albertini E, Raggi L, Gigliotti G, Bencivenga M (2010) Ecologia di *Tuber magnatum* Pico nell'alta valle del Chiascio. In: Donnini D (ed) Proceedings of the 3rd international congress on *Tuber*: Comunità Montana dei Monti Martani Serano e Subasio, Spoleto

Douet JP, Castroviejo M, Mabru D, Chevalier G, Dupré C, Bergougnoux F, Ricard JM, Médina B (2004) Rapid molecular typing of *Tuber melanosporum*, *T. brumale* and *T. indicum* from tree seedlings and canned truffles. Anal Bioanal Chem 379:668–673. doi:10.1007/s00216-004-2643-9

El Karkouri K, Murat C, Zampieri E, Bonfante P (2007) Identification of internal transcribed Spacer sequence motifs in truffles: a first step toward their DNA barcoding. Appl Environ Microbiol 73:5320–5330. doi:10.1128/AEM.00530-07

Etayo ML, De Miguel AM (2001) Effect of mulching on *Tuber melanosporum* Vitt. mycorrhizae in a cultivated truffle bed *vs*. other competing mycorrhizae. In: Actes du Ve Congrès International: Science et Culture de la Truffe. Federation Française des Trufficulteurs, Paris

Fortas Z, Chevalier G (1992) Effet des conditions de culture sur la mycorhization de l'*Helianthemum guttatum* par trios especes de terfez des genres *Terfezia* et *Tirmania* d'Algerie. Can J Bot 70:2453–2460. doi:10.1139/b92-303

Frey-Klett P, Garbaye J, Tarkka M (2007) The mycorrhiza helper bacteria revisited. New Phytol 176:22–36. doi:10.1111/j.1469-8137.2007.02191.x

Garcia-Barreda S, Reyna S (2011) Below-ground ectomycorrhizal community in natural *Tuber melanosporum* truffle grounds and dynamics after canopy opening. Mycorrhiza. doi:10.1007/s00572-011-0410-2

Giambernardi TA, Rodeck U, Klebe RJ (1998) Bovine serum albumin reverses inhibition of RT-PCR by melanin. Biotechniques 25:564–566

Giomaro G, Zambonelli A, Sisti D, Cecchini M, Evangelista V, Stocchi V (2000) Anatomical and morphological characterization of mycorrhizas of five strain of *Tuber borchii* Vittad. Mycorrhiza 10:107–114. doi:10.1007/s005720000065

Giomaro G, Sisti D, Zambonelli A, Amicucci A, Cecchini M, Comandini O, Stocchi V (2002) Comparative and molecular characterization in *Tilia americana* and *Quercus pubescens* with *Tuber brumale*. FEMS Microbiol Lett 216:9–14. doi:10.1016/S0378-1097(02)00962-x

Giraud M (1988) Prélévement et analyse de mycorhizes. Bull. FNTP, vol 10. In: La Truffe. CTIFL, Paris, pp 49–63

Granetti B, Baciarelli Falini L (1997) Competizione tra le micorrize di *Tuber melanosporum* Vitt. e quelle di altri fungi in una tartufaia coltivata a *Quercus ilex* L. Micol Ital 3:45–59

Gryndler M, Hršelová H, Soukupová L, Streiblová E, Valda S, Borovička J, Gryndlerová H, Gažo J, Miko M (2011) Detection of summer truffle (*Tuber aestivum* Vittad.) in ectomycorrhizae and in soil using specific primers. FEMS Microbiol Lett 318:84–89. doi:10.1111/j.1574-6968.2011.02243.x

Hall I, Brown G, Zambonelli A (2007) Taming the truffle. The history, lore, and science of the ultimate mushroom. Timber, Portland, OR

Hall IR, Lyon A, Wang Y, Buchanan P (2011) A list of putative edible or medicinal ectomycorrhizal mushrooms (revised). Dunedin, Truffles and Mushrooms (Consulting) Limited. http://www.trufflesandmushrooms.co.nz/Hall%20et%20al.%20A%20putative%20List%20of%20EEMM%202011.pdf

Hortal S, Pera J, Parladé J (2008) Tracking mycorrhizas and extraradical mycelium of the edible fungus *Lactarius deliciosus* under field competition with *Rhizopogon* spp. Mycorrhiza 18:69–77. doi:10.1007/s00572-007-0160-3

Horton TR, Bruns TD (2001) The molecular revolution in ectomycorrhizal ecology: peeking into the black-box. Mol Ecol 10:855–1871. doi:10.1046/j.0962-1083.2001.01333.x

Illyés Z, Ouanphanivanh N, Rudnóy S, Orczán AK, Bratek Z (2010) The most recent results on orchid mycorrhizal fungi in Hungary. Acta Biol Hung 61(Suppl):68–76. doi:10.1556/ABiol.61.2010.Suppl.8

Iotti M, Zambonelli A (2006) A quick and precise technique for identifying ectomycorrhizas by PCR. Mycol Res 110:60–65. doi:10.1016/j.mycres.2005.09.010

Iotti M, Amicucci A, Bonito G, Bonuso E, Stocchi V, Zambonelli A (2007) Selection of a set of specific primers for the identification of *Tuber rufum*: a truffle species with high genetic variability. FEMS Microbiol Lett 277:223–231. doi:10.1111/j.1574-6968.2007.00963.x

Iotti M, Lancellotti E, Hall I, Zambonelli A (2010) The ectomycorrhizal community in natural *Tuber borchii* grounds. FEMS Microbiol Ecol 72:250–260. doi:10.1111/j.1574-6941.2010.00844.x

Iotti M, Leonardi M, Oddis M, Salerni E, Baraldi E, Zambonelli A (2012) Development and validation of a real-time PCR assay for detection and quantification of *Tuber magnatum* in soil. BMC Microbiol 12:93. doi:10.1186/1471-2180-12-93

Iwanski M, Rudawska M (2007) Ectomycorrhizal colonization of naturally regenerating *Pinus sylvestris* L. seedlings growing in different micro-habitats in boreal forest. Mycorrhiza 17:461–467. doi:10.1007/s00572-007-0132-7

Jumpponen A, Egerton-Warburton LM (2005) Mycorrhizal fungi in successional environments: a community assembly model incorporating host plant, environmental, and biotic filters. In: Dighton J, White JF, Oudemans P (eds) The fungal community: its organization and role in the ecosystem. Taylor & Francis, London

Kennedy P (2010) Ectomycorrhizal fungi and interspecific competition: species interactions, community structure, coexistence mechanisms, and future directions. New Phytol 187:895–910. doi:10.1111/j.1469-8137.2010.03399.x

Kõljalg U, Larsson KH, Abarenkov K, Nilsson RH, Alexander IJ, Eberhardt U, Erland S, Høiland K, Kjøller R, Larsson E, Pennanen T, Sen R, Taylor AFS, Tedersoo L, Vralstad T, Ursing BM (2005) UNITE: a database providing Web-based methods for the molecular identification of ectomycorrhizal fungi. New Phytol 166:1063–1068. doi:10.1111/j.1469-8137.2005.01376.x

Kreader CA (1996) Relief of amplification inhibition in PCR with bovine serum albumin or T4 gene 32 protein. Appl Environ Microbiol 62:1102–1106

Kunin V, Engelbrektson A, Ochman H, Hugenholtz P (2010) Wrinkles in the rare biosphere: pyrosequencing errors can lead to artificial inflation of diversity estimates. Environ Microbiol 12:118–123. doi:10.1111/j.1462-2920.2009.02051.x

Lancellotti E, Iotti M, Melis A, Zambonelli A, Franceschini A (2012) eMyCo—Ectomycorrhizal community database. IOBC/WPRS Bull 76:133–138

Landeweert R, Leeflang P, Kuyper TW, Hoffland E, Rosling A, Wernars K, Smit E (2003) Molecular identification of ectomycorrhizal mycelium in soil horizons. Appl Environ Microbiol 69:327–333. doi:10.1128/AEM.69.1.327-333.2003

Lian B, Zang J, Hou W, Yuan S, Smith DL (2008) PCR-based sensitive detection of the edible fungus *Boletus edulis* from rDNA ITS sequences. Electron J Biotechnol 11. doi:10.2225/vol11-issue3-fulltext-4

Lindahl B, Stenlid J, Olsson S, Finlay R (2001) Effects of resource availability on mycelial interactions and 32P transfer between a saprotrophic and an ectomycorrhizal fungus in soil microcosms. FEMS Microbiol Ecol 38:43–52. doi:10.1111/j.1574-6941.2001.tb00880.x

Liu KL, Porras-Alfaro A, Kuske CR, Eichorst SA, Xie G (2011) Accurate, rapid taxonomic classification of fungal large subunit rRNA genes. Appl Environ Microbiol 78:1523–1533. doi:10.1128/ AEM.06826-11

Martin F, Kohler A, Duplessis S (2007) Living in harmony in the wood underground: ectomycorrhizal genomics. Curr Opin Plant Biol 10:204–210. doi:10.1016/j.pbi.2007.01.006

Mello A, Garnero L, Bonfante P (1999) Specific PCR-primers as a reliable tool for the detection of white truffles in mycorrhizal roots. New Phytol 14:511–516. doi:10.1046/j.1469-8137.1999.00356.x

Mello A, Fontana A, Meotto F, Comandini O, Bonfante P (2001) Molecular and morphological characterization of *Tuber magnatum* mycorrhizas in a long-term survey. Microbiol Res 155:279–284

Mello A, Cantisani A, Vizzini A, Bonfante P (2002) Genetic variability of *Tuber uncinatum* and its relatedness to other black truffles. Environ Microbiol 4:584–594. doi:10.1046/j.1462-2920.2002.00343.x

Mello A, Ghignone S, Vizzini A, Sechi C, Ruiu P, Bonfante P (2006) ITS primers for the identification of marketable Boletus. J Biotechnol 121:318–329. doi:10.1016/j.jbiotec.2005.08.022

Mello A, Miozzi L, Vizzini A, Napoli C, Kowalchuk G, Bonfante P (2010) Bacterial and fungal communities associated with *Tuber magnatum* productive niches. Plant Biosyst 144:323–332. doi:10.1080/11263500903374724

Mello A, Napoli C, Murat C, Morin E, Marceddu G, Bonfante P (2011) ITS-1 versus ITS-2 pyrosequencing: a comparison of fungal populations in truffle-grounds. Mycologia 103:1184–1193. doi:10.3852/11-027

Murat C, Vizzini A, Bonfante P, Mello A (2005) Morphological and molecular typing of the below-ground fungal community in a natural *Tuber magnatum* truffle-ground. FEMS Microbiol Lett 245:307–313. doi:10.1016/j.femsle.2005.03.019

Napoli C, Mello A, Bonfante P (2008) Dissecting the rhizosphere complexity: the truffle-ground study case. Rend Lincei 19:241–259. doi:10.1007/s12210-008-0017-5

Napoli C, Mello A, Borra A, Vizzini A, Sourzat P, Bonfante P (2010) *Tuber melanosporum*, when dominant, affects fungal dynamics in truffle grounds. New Phytol 185:237–247. doi:10.1111/j.1469-8137.2009.03053.x

Nilsson RH, Ryberg M, Kristiansson E, Abarenkov K, Larsson K-H, Kõljalg U (2006) Taxonomic reliability of DNA sequences in public sequence databases: a fungal perspective. PLoS One 1: e59. doi:10.1371/journal.pone.0000059

Nilsson RH, Kristiansson E, Ryberg M, Hallenberg N, Larsson KH (2008) Intraspecific ITS variability in the kingdom fungi as expressed in the international sequence databases and its implications for molecular species identification. Evol Bioinform 4:193–201

Pacioni G (1991) Effects of *Tuber* metabolites on rhizospheric environment. Mycol Res 95:1355–1358

Parladé J, Hortal S, Pera J, Galipienso L (2007) Quantitative detection of *Lactarius deliciosus* extraradical soil mycelium by real-time PCR and its application in the study of fungal persistence and interspecific competition. J Biotechnol 128:14–23. doi:10.1016/j.jbiotec.2006.09.010

Paulitz TC, Linderman RG (1991) Mycorrhizal interactions with soil organisms. In: Arora DK, Rai B, Mukerji KG, Knudsen G (eds) Handbook of applied mycology. Soil and plants. Marcel Dekker, New York

Peay KG, Kennedy PG, Bruns TD (2008) Fungal community ecology: a hybrid beast with a molecular master. Bioscience 58:799–810

Peintner U, Iotti M, Klotz P, Bonuso E, Zambonelli A (2007) Soil fungal communities in a *Castanea sativa* (chestnut) forest producing large quantities of *Boletus edulis sensu lato* (porcini): where is the mycelium of porcini? Environ Microbiol 9:880–889. doi:10.1111/j.1462-2920.2006.01208.x

Pielou EC (1969) An introduction to mathematical ecology. Wiley, New York

Porter TM, Golding GB (2011) Are similarity- or phylogeny-based methods more appropriate for classifying internal transcribed spacer (ITS) metagenomic amplicons? New Phytol 192:775–782. doi:10.1111/j.1469-8137.2011.03838.x

Rambold G, Agerer R (1997) DEEMY—the concept of a characterization and determination system for ectomycorrhizae. Mycorrhiza 7:113–116

Reich M, Kohler A, Martin F, Buée M (2009) Development and validation of an oligonucleotide microarray to characterise ectomycorrhizal fungal communities. BMC Microbiol 9:241. doi:10.1186/1471-2180-9-241

Rinaldi AC, Comadini O, Kuyper TW (2008) Ectomycorrhizal fungal diversity: separating the wheat from the chaff. Fungal Divers 33:1–45

Rubini A, Paolocci F, Granetti B, Arcioni S (1998) Single step molecular characterization of morphologically similar black truffle species. FEMS Microbiol Lett 164:7–12. doi:10.1111/j.1574-6968.1998.tb13060.x

Rubini A, Paolocci F, Granetti B, Arcioni S (2001) Morphological characterization of molecular-typed *Tuber magnatum* ectomycorrhizae. Mycorrhiza 11:179–185. doi:10.1007/s005720100116

Rubini A, Belfiori B, Passeri V, Baciarelli Falini L, Arcioni S, Riccioni C, Paolocci F (2011) The AD-type ectomycorrhizas, one of the most common morphotypes present in truffle fields,

result from fungi belonging to the *Trichophaea woolhopeia* species complex. Mycorrhiza 21:17–25. doi:10.1007/s00572-010-0308-4

Sejalon-Delmas N, Roux C, Martins M, Kulifaj M, Becard G, Dargent R (2000) Molecular tools for the identification of *Tuber melanosporum* in agroindustry. J Agric Food Chem 48:2608–2613. doi:10.1021/jf9910382

Selosse MA, Faccio A, Scappaticci G, Bonfante P (2004) Chlorophyllous and achlorophyllous specimens of *Epipactis microphylla* (Neottieae, Orchidaceae) are associated with ectomycorrhizal septomycetes, including truffles. Microb Ecol 47:416–426. doi:10.1007/s00248-003-2034-3

Shannon CE, Weaver A (1949) The mathematical theory of communication. University of Illinois Press, Urbana, IL

Sisti D, Giomaro G, Cecchini M, Faccio A, Novero M, Bonfante P (2003) Two genetically related strains of *Tuber borchii* produce *Tilia* mycorrhizas with different morphological traits. Mycorrhiza 13:107–115. doi:10.1007/s00572-002-0220-7

Smith SE, Read DJ (2008) Mycorrhizal symbiosis, 3rd edn. Academic, London

Smith ME, Douhan GW, Rizzo DM (2007) Ectomycorrhizal community structure in a xeric *Quercus* woodland based on rDNA sequence analysis of sporocarps and pooled roots. New Phytol 174:847–863. doi:10.1111/j.1469-8137.2007.02040.x

Splivallo R, Novero M, Bertea CM, Bossi S, Bonfante P (2007) Truffle volatiles inhibit growth and induce an oxidative burst in *Arabidopsis thaliana*. New Phytol 175:417–424. doi:10.1111/j.1469-8137.2007.02141.x

Streiblová E, Gryndlerová H, Gryndler M (2012) Truffle brûlé: an efficient fungal life strategy. FEMS Microbiol Ecol. doi:10.1111/j.1574-6941.2011.01283.x

Suz LM, Martɪn MP, Colinas C (2006) Detection of *Tuber melanosporum* DNA in soil. FEMS Microbiol Lett 254:251–257. doi:10.1111/j.1574-6968.2005.00029.x

Suz LM, Martin MP, Oliach D, Fischer CR, Colinas C (2008) Mycelial abundance and other factors related to truffle productivity in *Tuber melanosporum-Quercus ilex* orchards. FEMS Microbiol Lett 285:72–78. doi:10.1111/j.1574-6968.2008.01213.x

Tedersoo L, Nilsson RH, Abarenkov K, Jairus T, Sadam A, Saar I, Bahram M, Bechem E, Chuyong G, Kõljalg U (2010) 454 Pyrosequencing and Sanger sequencing of tropical mycorrhizal fungi provide similar results but reveal substantial methodological biases. New Phytol 188:291–301. doi:10.1111/j.1469-8137.2010.03373

Tedersoo L, Abarenkov K, Nilsson RH, Schüßler A, Grelet G-A, Kohout P, Oja J, Bonito GM, Veldre V, Jairus T, Kõljalg U (2011) Tidying up international nucleotide sequence databases: ecological, geographical and sequence quality annotation of ITS sequences of mycorrhizal fungi. PLoS One 6:e24940. doi:10.1371/journal.pone.0024940

Tedersoo L, Bahram M, Toots M, Diédhiou AG, Henkel TW, Kjøller R, Morris MH, Nara K, Nouhra E, Peay K, Põlme S, Ryberg M, Smith ME, Kõljalg U (2012) Towards global patterns in the diversity and community structure of ectomycorrhizal fungi. Mol Ecol. doi:10.1111/j.1365-294X.2012.05602.x

Whittaker RH (1965) Dominance and diversity in land plant communities. Science 147:250–260. doi:10.1126/science.147.3655.250

Zambonelli A, Salomoni S, Pisi A (1993) Caratterizzazione anatomo-morfologica e micromorfologica delle micorrize di *Tuber* spp. su *Quercus pubescens* Willd. Micol Ital 3:73–90

Zambonelli A, Penjor D, Pisi A (1995) Effetto del triadimefon sulle micorrize di *Tuber borchii* Vitt. e di *Hebeloma sinapizans* (Paulet). Gill. Micol Ital 14:65–73

Zambonelli A, Pisi A, Tibiletti A (1997) Caratterizzazione anatomo-morfologica delle micorrize di. *Tuber indicum* Cooke & Masee su *Pinus pinea* L. e *Quercus cerris* L. Micol Ital 1:29–36

Zambonelli A, Iotti M, Amicucci PA (1999) Caratterizzazione anatomo-morfologica delle micorrize di *Tuber maculatum* Vittad. su *Ostrya carpinifolia* Scop. Micol Ital 3:29–35

Zambonelli A, Iotti M, Rossi I, Hall I (2000a) Interaction between *Tuber borchii* and other ectomycorrhizal fungi in a field plantation. Mycol Res 104:698–700. doi:10.1017/S0953756299001811

Zambonelli A, Rivetti C, Percurdani R, Ottonello S (2000b) TuberKey: a DELTA-based tool for the description and interactive identification of truffles. Mycotaxon 74:57–76

Zambonelli A, Iotti M, Zinoni F, Dallavalle E, Hall R (2005) Effect of mulching on *Tuber uncinatum* ectomycorrhizas in an experimental truffière. NZ J Crop Hortic 33:65–73. doi:10.1080/01140671.2005.9514332

Zambonelli A, Iotti M, Barbieri E, Amicucci A, Stocchi V, Peintner U, Hall I (2009) The microbial communities and fruiting of edible ectomycorrhizal mushrooms. Yunnan Zhiwu Yanjiu IWEMM5 16:81–85

Zambonelli A, Iotti M, Bonuso E, Garbellotto M, Hall IR (2010) Taxonomic and quality issues associated with the commercialization of *Tuber borchii*. In: Donnini D (ed) Atti del 3° Congresso Internazionale di Spoleto sul Tartufo, Spoleto, 25–28 Nov 2008. Comunità Montana dei Monti Martani, Serano e Subasio, pp 295–299

Zampieri E, Murat C, Cagnasso M, Bonfante P, Mello A (2010) Soil analysis reveals the presence of an extended mycelial network in a *Tuber magnatum* truffle-ground. FEMS Microbiol Ecol 71:43–49. doi:10.1111/j.1574-6941.2009.00783.x